

# Homework 11

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## Problem 11.1.

### Solution 11.1.

1.  $\det(2A) = 2^3 \det(A) = 40$ .
2.  $\det(-A) = (-1)^3 \det(A) = -5$ .
3.  $\det(A^2) = \det(A)^2 = 25$ .
4.  $\det(A^{-1}) = \det(A)^{-1} = \frac{1}{5}$ .
5.  $\det(A^T) = \det(A) = 5$ .
- 6.

### Method 1: Row operation

$$\begin{vmatrix} \mathbf{a}_1^T - \mathbf{a}_3^T \\ \mathbf{a}_2^T - \mathbf{a}_1^T \\ \mathbf{a}_3^T - \mathbf{a}_2^T \end{vmatrix} = \begin{vmatrix} \mathbf{a}_1^T - \mathbf{a}_3^T \\ \mathbf{a}_2^T - \mathbf{a}_1^T \\ \mathbf{a}_1^T - \mathbf{a}_2^T \end{vmatrix} = 0.$$

### Method 2:

$$\begin{aligned} \begin{bmatrix} \mathbf{a}_1^T - \mathbf{a}_3^T \\ \mathbf{a}_2^T - \mathbf{a}_1^T \\ \mathbf{a}_3^T - \mathbf{a}_2^T \end{bmatrix} &= \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \mathbf{a}_3^T \end{bmatrix} - \begin{bmatrix} \mathbf{a}_3^T \\ \mathbf{a}_1^T \\ \mathbf{a}_2^T \end{bmatrix} \\ &= \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \mathbf{a}_3^T \end{bmatrix} - \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \mathbf{a}_3^T \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \mathbf{a}_3^T \end{bmatrix} \end{aligned}$$

Since  $\begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$  is not invertible, so the determinant is 0.

7.

### Method 1: Row operation

$$\begin{vmatrix} \mathbf{a}_1^T + \mathbf{a}_3^T \\ \mathbf{a}_2^T + \mathbf{a}_1^T \\ \mathbf{a}_3^T + \mathbf{a}_2^T \end{vmatrix} = \begin{vmatrix} \mathbf{a}_1^T + \mathbf{a}_3^T \\ \mathbf{a}_2^T - \mathbf{a}_3^T \\ \mathbf{a}_3^T + \mathbf{a}_2^T \end{vmatrix} = \begin{vmatrix} \mathbf{a}_1^T + \mathbf{a}_3^T \\ \mathbf{a}_2^T - \mathbf{a}_3^T \\ 2\mathbf{a}_3^T \end{vmatrix} = 2 \begin{vmatrix} \mathbf{a}_1^T + \mathbf{a}_3^T \\ \mathbf{a}_2^T - \mathbf{a}_3^T \\ \mathbf{a}_3^T \end{vmatrix} = 2 \begin{vmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \mathbf{a}_3^T \end{vmatrix} = 10.$$

### Method 2:

Similar to 6,  $\begin{bmatrix} \mathbf{a}_1^T + \mathbf{a}_3^T \\ \mathbf{a}_2^T + \mathbf{a}_1^T \\ \mathbf{a}_3^T + \mathbf{a}_2^T \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \mathbf{a}_3^T \end{bmatrix}$ . So  $\begin{bmatrix} \mathbf{a}_1^T + \mathbf{a}_3^T \\ \mathbf{a}_2^T + \mathbf{a}_1^T \\ \mathbf{a}_3^T + \mathbf{a}_2^T \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \det(A) = 10$ .

## Problem 11.2.

### Solution 11.2.

1. Proof:

$A = QR \implies [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n] = Q [\mathbf{r}_1 \ \mathbf{r}_2 \ \cdots \ \mathbf{r}_n] \implies [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n] = [Q\mathbf{r}_1 \ Q\mathbf{r}_2 \ \cdots \ Q\mathbf{r}_n] \implies \mathbf{a}_i = Q\mathbf{r}_i, 1 \leq i \leq n$ . But  $Q$  preserves length, so  $\|\mathbf{r}_i\| = \|\mathbf{a}_i\|$ .

2. Proof:

Set  $R = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ & r_{22} & & \vdots \\ & & \ddots & \vdots \\ & & & r_{nn} \end{bmatrix}$ , then  $\|\mathbf{r}_i\| = \sqrt{r_{1i}^2 + r_{2i}^2 + \cdots + r_{ii}^2}$ . So

$$\det(R) = r_{11}r_{22} \cdots r_{nn} \leq \sqrt{r_{11}^2} \sqrt{r_{12}^2 + r_{22}^2} \cdots \sqrt{r_{1n}^2 + r_{2n}^2 + \cdots + r_{nn}^2} = \|\mathbf{r}_1\| \|\mathbf{r}_2\| \cdots \|\mathbf{r}_n\|.$$

3. Proof:

$$|\det(A)| = |\det(Q)\det(R)| = \det(R) \leq \|r_1\| \|r_2\| \cdots \|r_n\| = \|a_1\| \|a_2\| \cdots \|a_n\|.$$

**Problem 11.3.**

**Solution 11.3.**

1. False.

Set  $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ ,  $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ , then  $AB - BA = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ . The determinant is not 0.

2. False.

Set  $A$  is an  $n \times n$  matrix, then  $\det(-A) = (-1)^n \det(A)$ .

3. Proof:

Since  $n$  is odd,

$$A^T = -A \implies \det(A^T) = \det(-A) \implies \det(A) = -\det(A) \implies \det(A) = 0$$

So  $A$  is not invertible.

4. Proof:

Set  $L_i$  be the upper left  $i \times i$  block of  $L$ ,  $D_i$  be the upper left  $i \times i$  block of  $D$ ,  $U_i$  be the upper left  $i \times i$  block of  $U$ , then

$$A = \begin{bmatrix} A_i & * \\ * & * \end{bmatrix} = \begin{bmatrix} L_i & \\ * & * \end{bmatrix} \begin{bmatrix} D_i & \\ & * \end{bmatrix} \begin{bmatrix} U_i & * \\ & * \end{bmatrix}$$

where  $L_i$  is a unit lower triangular matrix,  $D_i$  is a diagonal matrix,  $U_i$  is a unit upper triangular matrix, and  $*$  is some matrices that we don't know and don't care.

By block matrix multiplication, we have  $A_i = L_i D_i U_i$ . Set  $D_i = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_i \end{bmatrix}$ , then

$$\det(A_i) = \det(L_i D_i U_i) = \det(L_i) \det(D_i) \det(U_i) = \det(D_i) = d_1 d_2 \cdots d_i$$

Similarly, we also have

$$\det(A_{i-1}) = \det(D_{i-1}) = d_1 d_2 \cdots d_{i-1}$$

$$\text{Hence } d_i = \frac{\det(A_i)}{\det(A_{i-1})}.$$

**Problem 11.4.**

**Solution 11.4.**

1. Determinant should be a real number, but  $AD - BC$  is a matrix. In fact, we have,

$$\begin{vmatrix} A & B \\ C & D \end{vmatrix} = \begin{vmatrix} A & B \\ O & D - CA^{-1}B \end{vmatrix} = \det(A) \det(D - CA^{-1}B).$$

2.

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1}\right) = \det\left(\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\right) = \frac{1}{(ad-bc)^2} \det\left(\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\right) = \frac{1}{ad-bc}.$$

3.  $\det(-BA) = -\det(BA) = -\det(B)\det(A)$  if and only if  $n$  is an odd number.

**Problem 11.5.**

**Solution 11.5.**

1. In the big formula, we pick entries as follows.

$$\begin{bmatrix} 2x & x & 1 & 2 \\ 1 & x & 1 & -1 \\ 3 & 2 & x & 1 \\ 1 & 1 & 1 & x \end{bmatrix}$$

This is the only possibility to form  $x^4$ . So the coefficient of  $x^4$  is 2.  
For  $x^3$ , we can pick entries as follows.

$$\begin{bmatrix} 2x & x & 1 & 2 \\ 1 & x & 1 & -1 \\ 3 & 2 & x & 1 \\ 1 & 1 & 1 & x \end{bmatrix}$$

This is the only possibility to form  $x^3$ . So the coefficient of  $x^3$  is  $-1$ .

2. Set  $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3 \ \mathbf{a}_4]$ . After multiplication,  $A = [\mathbf{a}_1 \ 2\mathbf{a}_2 \ 3\mathbf{a}_3 \ 4\mathbf{a}_4]$ . Since each term has one number from  $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4$ , respectively, each term will multiply  $1 \times 2 \times 3 \times 4 = 24$ . And  $\det(A)$  also  $\times 24$ .
3. By the Big Formula,

$$\det(A) = acfh - agfd - ecbh + egbd = (af - eb)(ch - gd).$$

### Problem 11.6.

#### Solution 11.6.

1.  $L_n$  and  $U_n$  are unit triangular matrix, so  $\det(P_n) = \det(L_n U_n) = \det(L_n) \det(U_n) = 1$ .
- 2.

#### Method 1: LU decomposition

The LU decomposition of  $P_n$  is  $P_n = L_n U_n$ , where  $U_n$  is the REF of  $P_n$ . In LU decomposition, we only use upper rows to kill the lower rows. After reducing the lower right entry of  $P_n$  by 1, the row operations to kill the lower rows remain the same, only lower right entry of REF of  $P_n$  decreases by 1. So  $L_n$  doesn't change, the lower right entry of  $U_n$  decreases by 1. However,  $U_n$  is a unit upper triangular matrix, it is not invertible after its lower right entry decreasing by 1. Hence  $\det(P_n) = 0$ .

#### Method 2: Multilinearity

Set  $P_n = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix}$ . By multilinearity of determinant, we have

$$\begin{aligned} \begin{vmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} - 1 \end{vmatrix} &= \begin{vmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{vmatrix} - \begin{vmatrix} p_{11} & p_{12} & \cdots & p_{1,n-1} & 0 \\ p_{21} & p_{22} & \cdots & p_{2,n-1} & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ p_{n-1,1} & p_{n-1,2} & \cdots & p_{n-1,n-1} & 0 \\ p_{n1} & p_{n2} & \cdots & p_{n,n-1} & 1 \end{vmatrix} \\ &= \det(P_n) - \det(P_{n-1}) \\ &= 1 - 1 \\ &= 0. \end{aligned}$$

#### Method 3: Cofactor expansion

Set  $P_n = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix}$ , the cofactor of the lower right entry  $p_{nn}$  is

$$c_{nn} = (-1)^{n+n} \det(P_{n-1}) = 1.$$

So if  $p_{nn}$  decreases by 1, then  $\det(P_n)$  also reduces by 1. Hence the determinant is 0.

### Problem 11.7.

#### Solution 11.7.

- 1.

$$\det(H_n) = \begin{vmatrix} 2 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 1 & \cdots & 1 \\ & & 1 & \ddots & \vdots \\ & & & \ddots & 1 \\ & & & & 1 & 2 \end{vmatrix}_{n \times n}$$

Expansion by the first column, we have

$$\begin{aligned}
\begin{vmatrix} 2 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 1 & \cdots & 1 \\ & 1 & \ddots & \ddots & \vdots \\ & & \ddots & \ddots & 1 \\ & & & 1 & 2 \end{vmatrix}_{n \times n} &= 2 \begin{vmatrix} 2 & 1 & \cdots & 1 \\ 1 & \ddots & \ddots & \vdots \\ & \ddots & \ddots & 1 \\ & & 1 & 2 \end{vmatrix}_{(n-1) \times (n-1)} - \begin{vmatrix} 1 & 1 & \cdots & 1 \\ 1 & 2 & \ddots & \vdots \\ & \ddots & \ddots & 1 \\ & & 1 & 2 \end{vmatrix}_{(n-1) \times (n-1)} \\
&= 2 \det(H_{n-1}) - \begin{vmatrix} 1 & 1 & \cdots & 1 \\ 1 & 2 & \ddots & \vdots \\ & \ddots & \ddots & 1 \\ & & 1 & 2 \end{vmatrix}_{(n-1) \times (n-1)}
\end{aligned}$$

By multilinearity of determinant,

$$\begin{aligned}
\begin{vmatrix} 1 & 1 & \cdots & 1 \\ 1 & 2 & \ddots & \vdots \\ & \ddots & \ddots & 1 \\ & & 1 & 2 \end{vmatrix}_{(n-1) \times (n-1)} &= \begin{vmatrix} 2 & 1 & \cdots & 1 \\ 1 & 2 & \ddots & \vdots \\ & \ddots & \ddots & 1 \\ & & 1 & 2 \end{vmatrix}_{(n-1) \times (n-1)} - \begin{vmatrix} 1 & 1 & \cdots & 1 \\ 0 & 2 & \ddots & \vdots \\ & 1 & \ddots & \ddots \\ & & \ddots & \ddots & 1 \\ & & & 1 & 2 \end{vmatrix}_{(n-1) \times (n-1)} \\
&= \begin{vmatrix} 2 & 1 & \cdots & 1 \\ 1 & 2 & \ddots & \vdots \\ & \ddots & \ddots & 1 \\ & & 1 & 2 \end{vmatrix}_{(n-1) \times (n-1)} - \begin{vmatrix} 2 & 1 & \cdots & 1 \\ 1 & 2 & \ddots & \vdots \\ & \ddots & \ddots & 1 \\ & & 1 & 2 \end{vmatrix}_{(n-2) \times (n-2)} \\
&= \det(H_{n-1}) - \det(H_{n-2})
\end{aligned}$$

Hence

$$\begin{aligned}
\det(H_n) &= 2 \det(H_{n-1}) - (\det(H_{n-1}) - \det(H_{n-2})) \\
&= \det(H_{n-1}) + \det(H_{n-2})
\end{aligned}$$

And  $\det(H_1) = 2, \det(H_2) = 3$ , so  $\det(H_n)$  is the Fibonacci sequence.

2.

$$\det(S_n) = \begin{vmatrix} 3 & 1 & & \\ 1 & 3 & \ddots & \\ & \ddots & \ddots & 1 \\ & & 1 & 3 \end{vmatrix}$$

Expansion by the first column, we have

$$\begin{aligned}
\begin{vmatrix} 3 & 1 & & \\ 1 & 3 & \ddots & \\ & \ddots & \ddots & 1 \\ & & 1 & 3 \end{vmatrix}_{n \times n} &= 3 \begin{vmatrix} 3 & 1 & & \\ 1 & 3 & \ddots & \\ & \ddots & \ddots & 1 \\ & & 1 & 3 \end{vmatrix}_{(n-1) \times (n-1)} - \begin{vmatrix} 1 & 3 & 1 & \\ 1 & 3 & 1 & \ddots \\ & \ddots & \ddots & \ddots \\ & & \ddots & \ddots & 1 \\ & & & 1 & 3 \end{vmatrix}_{(n-1) \times (n-1)} \\
&= 3 \det(S_{n-1}) - \begin{vmatrix} 3 & 1 & & \\ 1 & \ddots & \ddots & \\ & \ddots & \ddots & 1 \\ & & 1 & 3 \end{vmatrix}_{(n-2) \times (n-2)} \\
&= 3 \det(S_{n-1}) - \det(S_{n-2})
\end{aligned}$$

$\det(S_n)$  is the  $2n + 2$  term of the Fibonacci sequence.